Naive Lie Theory 3.7 Exercises

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Notes

I have to say there is a more beautiful solution to 3.7.4. But due to limited space I omitted it. It's an inductive argument. For the base case $SU(1)$ is trivially connected.

Assume $SU(n-1)$ is path-connected. There is a smooth surjection $\pi : SU(n) \to S^{2n-1}$ where $\pi(A)$ is the first column of A. Fix a vector $v \in S^{2n-1}$,

$$\pi^{-1}(v) = \{\begin{pmatrix} 1 & 0 \\ 0 & A \end{pmatrix} | A \in SU(n-1)\}$$

So $\pi^{-1}(v) \cong SU(n-1)$. This is actually a fiber bundle

$$SU(n-1) \to SU(n) \to S^{2n-1}$$

However, one might ask what a fiber bundle is. Just think of it as a "local" product space, where $SU(n)$ is locally a product of $SU(n-1)$ and S^{2n-1}, but not necessarily an exact product space globally. An example will be the Mobius bundle, which locally looks like $\mathbb{R} \times S^1$, and the Hopf fibration, which appears a lot in previous chapters.

Anyway, back to our problem, if the base (S^{2n-1}) and fiber ($SU(n-1)$) are both path-connected, then so is the total space. To be more specific, take $x, y \in SU(n)$. As S^{2n-1} is path-connected, we can move from $\pi(x)$ to $\pi(y)$ by some $\gamma(t) : I \to S^{2n-1}$. It follows a lifted map $\bar{\gamma} : I \to SU(n)$ such that $\pi(\bar{\gamma}(t)) = \gamma(t)$. Now the problem is that $\bar{\gamma}(0) = x$ but $\bar{\gamma}(1) \in \pi^{-1}(\pi(y))$. However, $\pi^{-1}(\pi(y)) \cong SU(n-1)$ and given $SU(n-1)$ is path-connected, we can have another path inside $\pi^{-1}(\pi(y))$, $\delta : I \to \pi^{-1}(\pi(y))$ with $\delta(0) = \bar{\gamma}(1)$ and $\delta(0) = y$. x first moves along $\bar{\gamma}$ then δ to y. Great.

Although this is a more interesting and less machinery solution than what I actually put under 3.7.4, it requires significantly more topological prerequisite.

3.7.1

Check chapter 1.3. Quite easy to check that $Z(G) = \{\pm 1\}$.

3.7.2

Check the homomorphism $\det : U(n) \to S^1$. Its kernel is just $SU(n)$. For any $A \in U(n)$ with $\det(A) = \alpha$, take $\lambda = \alpha^{\frac{1}{n}}$, and $(\lambda I)^{-1}A \in SU(n)$. Therefore, every $A \in U(n)$ can be written as

$$A = \lambda IS$$

with some $S \in SU(n)$. Hence we say

$$U(n) = Z(U(n))SU(n) = \{xy|x \in Z(U(n)), y \in SU(n)\}$$

However, the order doesn't matter, since anything in $Z(U(n))$ of course commute with the whole $SU(n)$. Moreover,

$$Z(U(n)) \cap SU(n) = Z(SU(n))$$

Consider a very natural homomorphism $\varphi : SU(n) \to U(n)/Z(U(n))$

$$\varphi(A) = [A]$$

where $[A] = AZ(U(n))$ is the coset.

If $\varphi(A) = [I]$ then $A \in Z(U(n))$. But since $A \in SU(n)$, then

$$A \in Z(U(n)) \cap SU(n) = Z(SU(n))$$

So the kernel of φ is exactly $Z(SU(n))$. One has to be careful here. Taking the quotient only gives an injective homomorphism, not isomorphism yet:

$$\bar{\varphi} : SU(n)/Z(SU(n)) \hookrightarrow U(n)/Z(U(n))$$

Finally check the surjectivity. Take any $[A] \in U(n)/Z(U(n))$ and λ such that $\det(\lambda^{-1}A) = 1$. Then $\lambda^{-1}A \in SU(n)$, and $[A] = [\lambda^{-1}A]$. Thus every coset has a representative in $SU(n)/Z(SU(n)$, and $\bar{\varphi}$ is surjective, and we are done.

3.7.3

Given $Z(SU(2)) = \{\pm I\}$, check chapter 1.5 and 2.3. It explicit says $SU(2)/\{\pm I\} = SO(3)$.

3.7.4

We can reduce the problem to $SU(n)$. For any $A \in U(n)$, say $\det(A) = e^{i\alpha} \in S^1$. Define the path

$$A_t = \lambda(t)A$$

where $\lambda(t) = e^{\frac{-it\alpha}{n}}$. Then $A_0 = A$ and $A_1 \in SU(n)$. We only have to check the path-connectedness of $SU(n)$.

Let $A \in SU(n)$. It is diagonalizable

$$A = BZ_{\theta_1 \ldots \theta_n} B^{-1}$$

for some $B \in U(n)$. Given $\det(A) = 1$, $\Sigma_{i=1}^n \theta_i \in 2\pi\mathbb{Z}$. Define the path

$$A_t = BZ_{t\theta_1 \ldots t\theta_n} B^{-1}$$

where $A_0 = I$ and $A_1 = A$. So every $A \in SU(n)$ is connected with I and thus $SU(n)$ is path-connected.